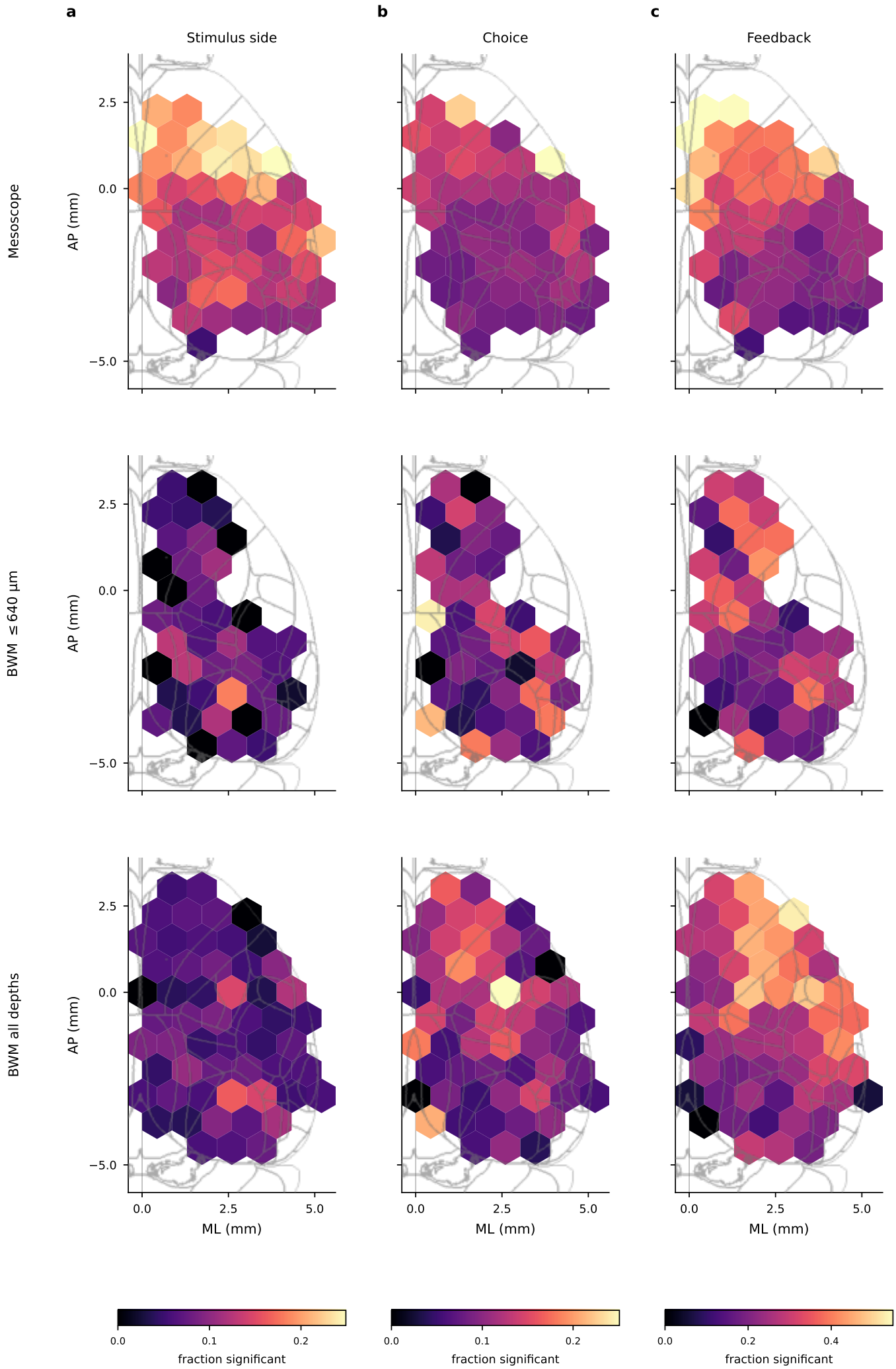


Figure 1: Tuning results: mesoscope two-photon imaging (top row; deconvolved activity of isocortex neurons) vs Brain Wide Map Neuropixels units within 640 μm of the pial surface (middle row; twice the depth range the mesoscope images, 174–318 μm) and at all depths (bottom row; reference). Each neuron is assigned to a hexagonal bin (1 mm vertex to vertex) by its ML and AP Allen CCF position; hemispheres folded onto the right. Grey lines: Allen top-view region boundaries. Colour scales are shared across rows. Single-neuron tuning: the BWM single-cell test (brainwidemap single_cell_stats, re-implemented on the GPU and checked against the original), applied identically to both modalities to the mean activity in each window. Mann–Whitney U statistics are computed within condition strata, summed and normalised; the null distribution comes from 3000 permutations of the condition labels within consecutive blocks, and p is the rank of the observed statistic. Mesoscope windows are 0.3 s long because the frame period is about 0.2 s. **a** Fraction of neurons per bin tuned to stimulus side ($p < 0.05$; bins with $\geq$10 neurons pooled across sessions); window $[0, 0.1]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to stimulus onset. **b** Choice; window $[-0.1, 0]$ s (BWM) or $[-0.3, 0]$ s (mesoscope) relative to movement onset. **c** Feedback; window $[0, 0.2]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to feedback.